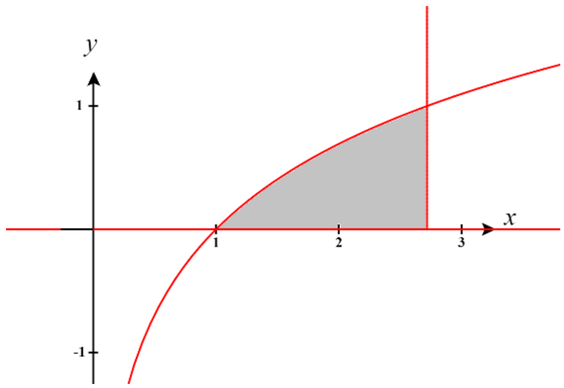
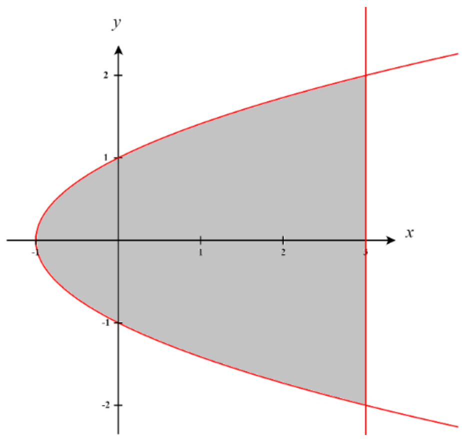
MTH 201 **Skills Check 1 Worksheet**

Skills checks cover pre-requisite material needed for the first unit of the course, areas & volumes. Answer the following questions neatly and completely. Calculators are not allowed.

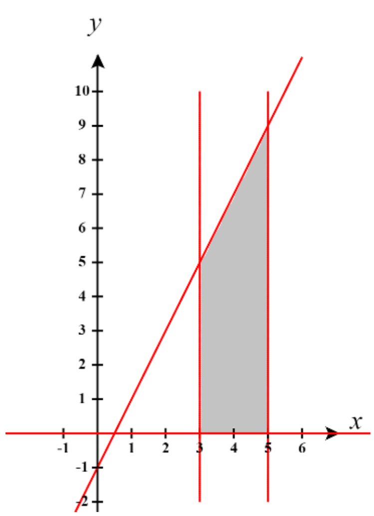
1. Shade the region defined by the following curves: $f(x) = \ln(x)$, $y = 0$, $x = e$.



2. Shade the region bounded by the following curves: $x = y^2 - 1$ & $x = 3$.



3. Shade the region bounded by the following curves: $f(x) = 2x - 1$, $x = 5$, $x = 3$, & $y = 0$.



4. Find the points of intersection of the curves: $x = 2y + 3$ & $x = y^2$.

   We find the point of intersection by setting the equations equal.

   We have

$$2y+3 = y^2 \;\rightarrow\; y^2 - 2y - 3 = 0 \;\rightarrow\; (y-3)(y+1) = 0 \;\rightarrow y = -1 \text{ or } 3$$

   These values of y correspond to the intersection points $(1,-1)$ & $(9,3)$.

5. Find the points of intersection of the curves: $y = x^3$ & $y = \sqrt{x}$.

   We find the point of intersection by setting the equations equal.

   We have

$$x^3 = \sqrt{x} \;\rightarrow\; x^3 - \sqrt{x} = 0 \;\rightarrow\; \sqrt{x}\left(x^{5/2} - 1\right) = 0 \;\rightarrow\; \sqrt{x} = 0 \text{ or } x^{5/2} - 1 = 0 \;\rightarrow\; x = 0 \text{ or } 1$$

   These values of x correspond to the intersection points $(0,0)$ & $(1,1)$.

6. Find the points of intersection of the curves: $y = x^2$, $y = 4 - x^2$.

   We find the point of intersection by setting the equations equal.

   We have

$$x^2 = 4 - x^2 \;\rightarrow\; 2x^2 = 4 \;\rightarrow\; x^2 = 2 \;\rightarrow\; x = \pm\sqrt{2}$$

   These values of x correspond to the intersection points $\left(-\sqrt{2},2\right)$ & $\left(\sqrt{2},2\right)$.

7. Evaluate the definite integral: $\int_{-3}^{-2}(x+3)^2\,dx$.

$$\int_{-3}^{-2}(x+3)^2\,dx=\int_{-3}^{-2}\left(x^2+6x+9\right)dx=\left(\frac{1}{3}x^3+3x^2+9x\right)\Bigg|_{-3}^{-2}=\left(\frac{-8}{3}+12-18\right)-\left(-9+27-27\right)=\left(-\frac{26}{3}\right)+(9)=\frac{1}{3}$$

8. Evaluate the definite integral: $\int_{2}^{5}\pi\left(\frac{1}{x}\right)^2dx$.

$$\int_{2}^{5}\pi\left(\frac{1}{x}\right)^2dx=\int_{2}^{5}\pi\frac{1}{x^2}\,dx=\int_{2}^{5}\pi x^{-2}dx=\left(-\pi x^{-1}\right)\Big|_2^5=\left(-\pi(5)^{-1}\right)-\left(-\pi(2)^{-1}\right)=\left(-\frac{\pi}{5}\right)+\left(\frac{\pi}{2}\right)=\frac{3\pi}{10}$$

9. Evaluate the definite integral: $\int_{1}^{2}\pi\left(k+\frac{1}{k}\right)dk$.

$$\int_{1}^{2}\pi\left(k+\frac{1}{k}\right)dk=\pi\int_{1}^{2}\left(k+\frac{1}{k}\right)dk=\pi\left(\frac{1}{2}k^2+\ln|k|\right)\Bigg|_1^2=\pi\left(\frac{1}{2}(2)^2+\ln(2)\right)-\pi\left(\frac{1}{2}(1)^2+\ln(1)\right).$$

$$=\left(2\pi+\pi\ln(2)\right)-\left(\frac{\pi}{2}\right)$$

$$=\frac{3\pi}{2}+\pi\ln(2)$$

10. Use a u-substitution to evaluate the definite integral: $\int_{-2}^{1} \frac{x}{2x^2+1}\,dx$.

Let $u = 2x^2+1$, then $du = 4x\,dx \;\rightarrow\; \frac{1}{4}du = xdx$.

Our bounds become $u(-2) = 2(-2)^2+1 = 9$ & $u(1) = 2(1)^2+1 = 3$.

So we have

$$\int_{-2}^{1} \frac{x}{2x^2+1}\,dx = \frac{1}{4}\int_{9}^{3} \frac{1}{u}\,du = \frac{1}{4}\ln|u|\Big|_9^3 = \frac{1}{4}\ln(3) - \frac{1}{4}\ln(9) = \frac{1}{4}\left[\ln(3) - \ln(9)\right] = \frac{1}{4}\ln\left(\frac{1}{3}\right)$$

11. Use a u-substitution to evaluate the definite integral: $\int_{0}^{\pi/3} 3\sin(x)e^{\cos(x)}dx$.

Let $u = \cos(x)$, then $du = -\sin(x)\,dx \;\rightarrow\; -du = \sin(x)\,dx$.

Our bounds become $u(0) = \cos(0) = 1$ & $u\left(\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$.

So we have

$$\int_{0}^{\pi/3} 3\sin(x)e^{\cos(x)}dx = -3\int_{1}^{1/2} e^u\,du = -3e^u\Big|_1^{1/2} = -3e^{1/2} + 3e^1 = 3\left[e - \sqrt{e}\right]$$

12. Use a u-substitution to evaluate the definite integral: $\int_{0}^{1} \pi t\left(t^2-1\right)^5 dt$.

Let $u = t^2-1$, then $du = 2t\,dt \;\rightarrow\; \frac{1}{2}du = tdt$.

Our bounds become $u(0) = (0)^2 - 1 = -1$ & $u(1) = (1)^2 - 1 = 0$.

So we have

$$\int_{0}^{1} \pi t\left(t^2-1\right)^5 dt = \frac{\pi}{2}\int_{-1}^{0} u^5\,du = \frac{\pi}{12}u^6\Big|_{-1}^{0} = \left(\frac{\pi}{12}(0)^6\right) - \left(\frac{\pi}{12}(-1)^6\right) = -\frac{\pi}{12}$$

13. Solve the equation for x: $y = \sqrt[3]{2x+1}$.

$$y = \sqrt[3]{2x+1} \quad \rightarrow \quad y^3 = 2x+1 \quad \rightarrow \quad x = \frac{y^3-1}{2}$$

14. Solve the equation for x: $y = 1 - \ln(x)$.

$$y = 1 - \ln(x) \quad \rightarrow \quad \ln(x) = 1 - y \quad \rightarrow \quad x = e^{1-y}$$

15. Solve the equation for x: $y = \dfrac{1}{2-x}$.

$$y = \frac{1}{2-x} \quad \rightarrow \quad 2 - x = \frac{1}{y} \quad \rightarrow \quad x = 2 - \frac{1}{y}$$